

A decorative graphic on the left side of the slide, consisting of a network of thin, light blue lines and small circles, resembling a circuit board or a neural network diagram. The lines are vertical and horizontal, with some diagonal connections, and the circles are placed at various points along these lines.

VERIFIABLE DECENTRALIZED TRAFFIC OPTIMIZATION USING EDGE SLMS

EBENEZER ISAAC

THE MULTI-BILLION DOLLAR STAGNATION

- **The Problem is Accelerating:** In 2025, congestion costs rose to **\$894 per driver** in the US (up from \$780 in 2024), with major cities like Chicago losing **112 hours** per commuter annually.
- **Legacy Tech is Failing:** 70% of global intersections still run on fixed-time plans or legacy adaptive systems (SCATS/SCOOT) rooted in **1980s centralized logic**.
- **The Latency Floor:** Current centralized systems have a hard latency floor (seconds to minutes) because data must travel from *Sensor > Cloud/Central Server > Decision > Traffic Light*.
- **Why It Matters:** We are trying to route 2026 autonomous vehicle traffic with 1985 telecommunications infrastructure. It's a bottleneck that cloud computing cannot solve due to latency.

THE "BLACK BOX" TRUST CRISIS

- **The Verifiability Trilemma:** Modern AI (Reinforcement Learning) is efficient but opaque. If an AI traffic light causes an accident, **who is liable?** The city? The vendor? The algorithm?
- **The "Hallucination" Risk:** Standard AI models can "hallucinate" or be tricked. A hacked billboard could fool a vision system into seeing a "Ghost Car," causing gridlock.
- **Current Solution Gap:** Competitors are rushing to deploy "Black Box" AI. Cities are rejecting them because they cannot audit *why* a light turned green.
- **The Market Opportunity:** The Smart Traffic Market is projected to hit **\$14.8B in 2025**, but the winner will be the *safest* system, not just the fastest.

OUR SOLUTION: THE "EDGE NEGOTIATOR" NODE

- **Architecture Shift:** Move the "Brain" from the Cloud to the Pole. Each intersection becomes an autonomous agent.
- **Semantic Reasoning:** Unlike SCATS (which sees "magnetic blips"), our system uses Vision-Language logic. It understands *"Ambulance blocked by truck"* vs. *"Normal rush hour."*
- **Performance Gains:** Research shows Decentralized Multi-Agent Reinforcement Learning (MARL) reduces travel time by **13%** and fuel consumption by **11%** compared to centralized systems.
- **The "Trust" Layer:** We wrap the AI inferences in a Decentralized Ledger. Every "Green Light" command is digitally signed and logged on a local ledger. We don't just say "Trust us"; we provide mathematical proof of safety.

CONCLUSION

- **Strategic Relevance:** This aligns with the global push for **Web 4.0 (DePIN)**—where physical infrastructure becomes intelligent and autonomous.
- **Economic Impact:** Reducing wait times by just 10% in a mid-sized city saves millions in lost productivity and fuel.
- **Our Competitive Moat:** While others build "faster" AI, we are building "accountable" AI. This is the feature that allows us to pass regulation.